

Building Mathematical Dispositions Posters

Three posters exploring brave mathematicians, divergent thinking, and the transformative power of questioning in math classrooms.

Source: Dance & Kaplan (2018) — Chapters Three, Four, and Five

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Dance & Kaplan (2018)

POSTER 01

Building Brave Mathematicians

Chapter Three — Dance & Kaplan (2018)

Building Brave Mathematicians

Mathematical confidence is not innate — it is cultivated through perseverance, productive mistakes, risk-taking, and self-reliance inside a classroom community built for growth.

BRAVERY

PERSEVERANCE

VALUING MISTAKES

RISK-TAKING

SELF-RELIANCE

GROWTH MINDSET

Being Brave in Mathematics

Chapter Three opens by asking which of three students — Yolanda, Irma, and Sabrina — learned the most from a challenging math problem. Irma is the answer: she struggled, tried multiple times, made mistakes along the way, and ultimately figured it out. In the Building Brave Mathematicians section of Chapter Three, Dance and Kaplan (2018) establish that it is the teacher's job to create a classroom community where all students have mathematical confidence and perseverance. Bravery in mathematics means returning to a problem after a setback, attempting an unfamiliar strategy, and believing that the struggle itself is the learning.

NARRATIVE • CHAPTER THREE • MATHEMATICAL CONFIDENCE: WHY IT MATTERS

In the Mathematical Confidence: Why It Matters section of Chapter Three, Dance and Kaplan (2018) explain that children with low self-confidence may get angry and give up when they feel they are not smart enough to figure a problem out — making it the classroom's responsibility to replace that fear with brave, supported persistence.

Perseverance

In the Perseverance section of Chapter Three, Dance and Kaplan (2018) note that children who possess mathematical confidence look at challenging problems in a completely different way — failure is a chance to learn and grow, a chance to reflect and ask what can be done better next time. They further explain that children with

Valuing Mistakes

In the Valuing Mistakes section of Chapter Three, Dance and Kaplan (2018) draw on brain research showing that when a student makes a mistake, synapses in the brain fire, forming new pathways and connections — and that new brain pathways are formed from making mistakes even when students do not know they are making them. Getting

mathematical confidence are able to persevere through challenging problems, trying again and again until they figure them out, and that it is the teacher's job to help students gain that perseverance.

NARRATIVE · CHAPTER THREE ·
PERSEVERANCE

In the Perseverance section of Chapter Three, Dance and Kaplan (2018) caution that students who define themselves as smart may actually avoid challenges to protect that identity — meaning the students who appear most confident are often the most at risk of giving up when real difficulty arrives.

the right answer, by contrast, forms no new pathways at all. Mistakes are not evidence of failure — they are the very mechanism through which mathematical understanding grows.

NARRATIVE · CHAPTER THREE ·
VALUING MISTAKES

In the Valuing Mistakes section of Chapter Three, Dance and Kaplan (2018) argue that in a math classroom, mistakes should be celebrated as great mistakes and learning opportunities, not looked at as simply wrong answers.

Children who possess mathematical confidence look at challenging math problems in a completely different way. Failure is a chance to learn and grow.

Dance & Kaplan (2018) · Chapter Three · Perseverance

Taking Risks

In the Taking Risks section of Chapter Three, Dance and Kaplan (2018) explain that students need to feel confident enough to take risks, try new strategies, and share their thinking even when it contradicts that of others. Risk-taking means trying a strategy that might not work, sharing a thought that might be wrong, and challenging a classmate's reasoning even when it feels uncomfortable. This confidence can only grow inside a classroom culture where

Self-Reliance

In the Self-Reliance section of Chapter Three, Dance and Kaplan (2018) describe how students who feel brave as mathematicians do not rely on the teacher to tell them what to do or how to solve a problem — instead they check their own work and justify their own thinking. Self-reliance is the culmination of all the other brave dispositions: when students persevere, value mistakes, and take risks consistently over time, they stop depending

wrong answers are treated as starting points, not endpoints.

on the teacher for confirmation and start trusting their own mathematical judgment.

Language That Builds Brave Mathematicians

In the Language Use section of Chapter Three, Dance and Kaplan (2018) identify two critical language practices. First, praise must target effort and process rather than intelligence — students praised for being smart begin to protect that identity by avoiding challenge. Second, the phrase I can't must be met with the single word yet. In that same section, Dance and Kaplan (2018) explain that when students were allowed to say I can't and teachers simply added yet to the end, students began to see success not as a two-path road where you get it or you don't, but as a bumpy, windy, hilly trail that can be pushed through.

NARRATIVE • CHAPTER THREE • LANGUAGE USE

In the Language Use section of Chapter Three, Dance and Kaplan (2018) explain that praising students for their intelligence causes them to begin to identify themselves as smart or not smart — and when students who identify as smart cannot figure out an answer, they face that failure as a threat to their identity rather than as a natural part of learning.

NARRATIVE • CHAPTER THREE • STUDENT LANGUAGE

In the Student Language section of Chapter Three, Dance and Kaplan (2018) also address the phrase This is easy, noting that a student who hears that phrase while struggling immediately feels less confident about their own skills and abilities — making the classroom culture as important as any instructional strategy.

Classroom Strategies for Building Brave Mathematicians

- In the Evaluating and Discussing Self-Confidence section of Chapter Three, Dance and Kaplan (2018) recommend administering a math attitude survey at the start of the year using a rating scale so students can honestly self-assess their feelings about math, with those results guiding the teacher's encouragement approach.
- In the Evaluating and Discussing Self-Confidence section of Chapter Three, Dance and Kaplan (2018) suggest connecting mathematics to skills students have already mastered through practice — such as riding a bike or playing a sport — so that struggle feels familiar and purposeful rather than alarming.

- In the Read-Alouds and Videos section of Chapter Three, Dance and Kaplan (2018) advocate using growth-mindset read-alouds and videos throughout the year to give students a story-based vocabulary for discussing perseverance and mistakes before they encounter those feelings in mathematics.

 - In the Perseverance in the Real World section of Chapter Three, Dance and Kaplan (2018) encourage sharing classroom stories of real-world perseverance — including the teacher's own struggles — so students understand that failure before success is universal, not personal.

 - In the Student Language section of Chapter Three, Dance and Kaplan (2018) call for replacing I can't with I can't yet so that limitations become temporary waypoints on a longer learning trail rather than fixed endpoints.
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Great Minds Think Differently

Chapter Four — Dance & Kaplan (2018)

Great Minds Think Differently

Mathematics is not a single road to a single answer. Students deepen understanding when they explore multiple strategies, justify their thinking, and connect their ideas with those of their peers.

MULTIPLE STRATEGIES

JUSTIFICATION

CLARIFYING THINKING

MAKING CONNECTIONS

REFLECTING

PROCESS OVER ANSWER

The Process of Getting to the Answer

In the opening of Chapter Four, Dance and Kaplan (2018) identify one of the biggest challenges in a problem-solving-based classroom as the belief that there is only one right way to do math. That same opening section establishes two key beliefs: first, that there are different strategies to solve a problem and students should be flexible in using a variety of them; and second, that while finding the answer is the end goal, it is more important to be able to explain how we got there. Without these beliefs in place, students will always prioritize speed over sense-making.

NARRATIVE • CHAPTER FOUR • MULTIPLE STRATEGIES AND JUSTIFICATION: WHY THEY MATTER

In the Multiple Strategies and Justification: Why They Matter section of Chapter Four, Dance and Kaplan (2018) explain that when students are able to explain their thinking and have freedom to pick a strategy that works for them, they are able to create meaning around mathematical concepts — meaning justification is not an add-on to mathematics but its core purpose.

Multiple Strategies

In the Multiple Strategies section of Chapter Four, Dance and Kaplan (2018) explain that when students are exposed to many strategies and representations, they learn to pick the strategy that not only makes the most sense but is also the most efficient given a particular problem. Comparing strategies also strengthens mathematical

Justification

In the Justification section of Chapter Four, Dance and Kaplan (2018) describe justification as the act of explaining not just what was done, but why. They note that when students focus on the process, they develop their understanding of mathematics because they are at the center of explaining, providing evidence, finding or creating

understanding — students who find the connections between a drawn-picture approach and a number-line approach understand both more deeply than students who used only one. Teachers must create a classroom culture that empowers students to bravely compare their strategies with those of their classmates.

examples, generalizing, analyzing, making predictions, applying concepts, representing ideas in different ways, and articulating connections. Justification makes thinking visible and gives teachers the evidence they need to assess understanding and catch misconceptions before they become habits.

Students gain a deeper understanding of numbers and mathematical concepts through the act of explaining their thinking because they are able to make sense of the problem on their own.

Dance & Kaplan (2018) • Chapter Four • Justification

Helping Students Clarify Their Thinking

In the Reflecting on Thinking section of Chapter Four, Dance and Kaplan (2018) encourage teachers to have students use sentence frames such as First I _____, next I _____, then I _____ to articulate each step they took toward a solution. Students are also asked to label every part of their models and strategies so that a partner or teacher should be able to look at a student's paper and understand exactly how they solved the problem. Clarity in mathematical thinking is a skill that must be explicitly taught and practiced, not assumed.

NARRATIVE • CHAPTER FOUR •
REFLECTING ON THINKING

In the Reflecting on Thinking section of Chapter Four, Dance and Kaplan (2018) state that students should be able to assess their own thinking so that they learn to choose more efficient

Making Connections Between Thinking

In the Student Shares section of Chapter Four, Dance and Kaplan (2018) explain that student shares allow learners to see similarities and differences between each other's strategies and make meaningful connections. When a student recognizes that a classmate's number-line approach reaches the same answer as their own drawn-picture approach, they deepen their understanding of both strategies — and of the underlying mathematical concept that makes both work. These peer-to-peer connections are among the most powerful learning moments a classroom can create.

strategies and develop more thorough justifications.

Reflecting on Thinking

In the Reading Thinking section of Chapter Four, Dance and Kaplan (2018) describe a technique in which teachers have students read thinking — examining a classmate's paper silently and trying to reconstruct the strategy used before any explanation is given. This reveals whether a student's work is truly clear and teaches students to evaluate the efficiency of their own strategy choices. In one classroom vignette from that same section of Chapter Four, Leo's paper could not be understood from the paper alone, and the class turned the moment into a community problem: what could Leo do next time to show his thinking more clearly?

NARRATIVE • CHAPTER FOUR • READING THINKING

In the Reading Thinking section of Chapter Four, Dance and Kaplan (2018) describe how, after the class worked through Leo's unclear paper, the teacher celebrated his work as a great mistake that allowed the class to learn — modeling for students that imperfect thinking shared publicly is more valuable than correct thinking kept private.

Classroom Structures That Support Great Thinking

- In the Tasks That Elicit Thinking section of Chapter Four, Dance and Kaplan (2018) recommend using tasks that require deeper thinking — such as asking students to explain who is correct in a math disagreement and why — rather than tasks that reward a single correct digit.
- In the Student Shares section of Chapter Four, Dance and Kaplan (2018) advocate for student shares where classmates explain each other's strategies in their own words, reinforcing active listening and deepening understanding of multiple approaches.
- In the Public Records section of Chapter Four, Dance and Kaplan (2018) encourage creating living anchor-chart posters that document students' strategies by name, giving them ownership and a daily reference for the classroom's growing mathematical thinking.
- In the Tasks That Elicit Thinking section of Chapter Four, Dance and Kaplan (2018) call for teachers to ask How do you know? and Why did you use that strategy? daily and repeatedly to shift the classroom's focus from the answer to the reasoning behind it.

→ In the Reflecting on Thinking section of Chapter Four, Dance and Kaplan (2018) expect words, numbers, and pictures or models to appear together on every student paper so that a reader can follow the full solution process without any additional explanation.

Thinking Through Questioning

Chapter Five — Dance & Kaplan (2018)

Thinking Through Questioning

Questions are among the most powerful tools a teacher possesses. Deliberate, targeted questioning transforms passive listeners into active mathematical thinkers who probe, justify, and extend their own reasoning.

CLARIFY & PROBE

GUIDE & EXTEND

ASSESS UNDERSTANDING

QUESTIONS VS STATEMENTS

TEACHER MODELING

TALK MOVES

NOTICING & WONDERING

The Power of Questions in Mathematics

In the Questioning: Why It Matters section of Chapter Five, Dance and Kaplan (2018) establish that being able to ask specific, targeted questions is a skill that must be learned and practiced by both the teacher and the students. As teachers question students, they are also modeling the exact questioning techniques students will eventually use with each other as they analyze and compare their classmates' mathematical reasoning. In the Types of Questions section of Chapter Five, Dance and Kaplan (2018) identify three categories of teacher questions — those that clarify and probe for justification, those that guide, challenge, and extend thinking, and those that assess understanding — and show how a skilled teacher moves fluidly between all three within a single conversation.

NARRATIVE • CHAPTER FIVE • TYPES OF QUESTIONS

In the Types of Questions section of Chapter Five, Dance and Kaplan (2018) explain that strategic, deliberate questions can push student thinking to deeper levels and open up opportunities for quick formative assessment — making questioning not a supplement to mathematics instruction but its backbone.

Questions That Clarify and Probe

In the Questions That Clarify and Probe Thinking section of Chapter Five, Dance and Kaplan (2018) explain that clarification questions help students understand a task,

Questions That Guide, Challenge, and Extend

In the Questions That Guide, Challenge, and Extend section of Chapter Five, Dance and Kaplan (2018) explain that these questions

make sense of a problem, and explain their thinking more clearly — and that probing questions encourage students to think about a problem at a deeper level. Both types are generally open-ended and should not give any clues or hints as to how to solve the problem. Together they reduce obstacles like reading comprehension barriers or unfamiliar vocabulary without removing the cognitive challenge of the mathematics itself.

NARRATIVE • CHAPTER FIVE •
QUESTIONS THAT CLARIFY AND
PROBE THINKING

In the Questions That Clarify and Probe Thinking section of Chapter Five, Dance and Kaplan (2018) note that clarifying and probing questions help decrease obstacles that get in the way of mathematical thinking, such as reading comprehension issues or a lack of language skills.

are used to push student thinking, often while creating cognitive dissonance, which increases the learning. They allow students to deepen their own reasoning as they justify their thinking, navigate through misconceptions, and help students who are stuck find an entry point into a problem. They also extend confident thinkers by changing the numbers or the context of the problem.

NARRATIVE • CHAPTER FIVE •
QUESTIONS THAT GUIDE,
CHALLENGE, AND EXTEND

In the Questions That Guide, Challenge, and Extend section of Chapter Five, Dance and Kaplan (2018) illustrate how Mrs. Dance extends a student's thinking by asking what would happen if the number of strawberries changed — using a single question to push fluency with harder numbers without changing the underlying mathematical concept.

Questions are one of the most valuable tools we have as teachers. Strategic, deliberate questions can push student thinking to deeper levels and open up opportunities for quick formative assessment.

Dance & Kaplan (2018) • Chapter Five • Types of Questions

Questions That Assess Understanding

In the Questions That Assess Understanding section of Chapter Five, Dance and Kaplan (2018) explain that questions used as formative assessment allow teachers to make real-time instructional decisions about what to revisit, who to challenge, and where misconceptions are hiding. In that section, Dance and Kaplan (2018) describe how Mrs. Dance asks students to place a thumb up, sideways, or down on their tummies before a whole-group share — a self-assessment signal that gives the teacher an

immediate classroom snapshot while protecting students from public embarrassment. In the same section, Dance and Kaplan (2018) recount how several students arrived at an incorrect answer by adding the two known numbers together — a common missing-part misconception that Mrs. Dance addressed entirely through questioning rather than direct correction.

Questions vs. Statements

When a teacher states the answer or explains the strategy, the cognitive load shifts from the student to the teacher. Questions preserve that load. In the Questions That Assess Understanding section of Chapter Five, Dance and Kaplan (2018) note that during the classroom vignette, Mrs. Dance simply facilitated conversation without explaining, justifying, or praising student ideas — returning all three of those responsibilities to the students where they belong. A question invites a student into thinking; a statement closes the door on it.

Teacher Talk Moves, Modeling, and Praise

In the Types of Questions section of Chapter Five, Dance and Kaplan (2018) emphasize that teachers must be flexible with their questioning because even with the best-planned lesson, a teacher can never quite know which direction a conversation may flow. In the Questioning: Why It Matters section of Chapter Five, Dance and Kaplan (2018) further note that becoming an artful questioner requires patience and practice and demands that teachers truly listen to students and be responsive to where they are in their understanding. Students who observe this patient, responsive modeling daily begin to replicate it in their own peer conversations.

CLARIFY & PROBE

- ? What do we know about this problem?
- ? What are we trying to figure out?
- ? Can you tell me more about that?
- ? What does that number mean in the problem?
- ? Can you explain that in your own words?

GUIDE & EXTEND

- ? Why did you use that strategy?
- ? Can you use a second strategy to prove it?
- ? What if the numbers were different?
- ? Which strategy was most efficient?
- ? Does that strategy always work?

ASSESS UNDERSTANDING

- ? Do you agree or disagree? Why?
- ? How is your strategy similar to ___'s?
- ? Can you explain ___'s strategy in your own words?
- ? Would someone thinking differently like to share?
- ? What could ___ do next time to show this more

clearly?

Question examples drawn from the *Types of Questions* section of Chapter Five in Dance & Kaplan (2018).

Noticing, Wondering, and the Role of Teacher Modeling

Noticing and wondering lower the barrier to participation by treating all observations as valid starting points — there is no wrong thing to notice. This approach gives every student permission to engage before they are certain, which is exactly the disposition that the brave mathematicians described in Chapter Three need. In the Questioning: Why It Matters section of Chapter Five, Dance and Kaplan (2018) emphasize that being an artful questioner requires patience and practice, and that the teacher must truly listen to students and be responsive to where they are in their understanding. In that same section, Dance and Kaplan (2018) note that as teachers question students, they are also modeling effective questioning techniques — meaning every question a teacher asks is simultaneously instruction and demonstration of what mathematical inquiry sounds like.

Reference

Dance, R., & Kaplan, T. (2018). *This is not a math book: Building mathematical dispositions in young learners*. Heinemann.

All citations draw exclusively from Chapters Three, Four, and Five of the source above.